Investigating the use of harmonic lasing to produce short-wavelength X-ray radiation in a Free Electron Laser

Ajay Mandlekar
Stanford Linear Accelerator Center

March 6, 2016

Mentors: Tor Raubenheimer, Juhao Wu (SLAC), Ryan Patterson (Caltech)

Abstract

The Linear Coherent Light Source (LCLS) produces high-power X-ray radiation that is used by scientists to obtain high-resolution images of samples in various applications. Now, with the prospect of building the LCLS-II, a more powerful successor to the LCLS, it may be possible to make the transition towards shorter wavelength and higher energy radiation. Harmonic lasing might be one way to realize these goals. This study investigates whether harmonic lasing will be a useful tool in the LCLS-II by comparing the generation of 21 keV radiation in the fundamental to the generation of 21 keV radiation as the third harmonic of 7 keV radiation. This study was conducted using Genesis simulation code and Matlab as a scripting tool to optimize various parameters and run several simulations. In optimizing the harmonic lasing case, it was important to understand the radiation generation mechanism and the transverse radiation profile and to optimize the transverse electron beam profile and the magnetic lattice. Future work will focus on verifying these findings and implementing them into the LCLS-II design.

1 Introduction

The Linear Coherent Light Source (LCLS), the world's most powerful X-ray laser, has enjoyed unprecedented success since its construction in 2009 [1] [2]. A Free Electron Laser (FEL), which generates x-rays by passing a high-energy electron beam through a long magnetic undulator, the LCLS is uniquely capable of producing X-rays that are both high in power and bunched in tiny pulses [1]. This property is critical for procuring high-quality images of atoms and molecuels before radiation damage occurs [1]. For example, the LCLS X-ray laser was recently used to investigate the role of a natural catalyst in photosynthesis [4]. Now, scientists at SLAC are working on designing and constructing the

LCLS-II, a more powerful successor to the LCLS.

The LCLS-II will include a number of important improvements over the LCLS-I including variable gap undulators with phase shifters between undulator segments [3]. This will allow much more flexibility in configuring the FEL since there is much more freedom in setting up the magnetic lattice that guides and focuses the beam, and raises the possibility of generating higher energy X-rays than are produced in the current LCLS [3]. Higher energy photons are of critical importance in many different experiments including those involving high-resolution diffraction, the spectroscopy of high mass elements (such as Zirconium), or conformational changes in disordered systems (liquids, gases). The conventional method of generation would be to tune the undulator to produce this radiation using the fundamental wavelength; however, generating high-energy photons has stringent requirements on the undulator and on the electron beam. The new capabilities of the LCLS-II offer the possibility of using an alternative method called harmonic lasing [7]. Through harmonic lasing, higher energy photons can be obtained from the undulator tuned to a fundamental at lower energies since the energy of the harmonic radiation that is generated is an integer multiple of the fundamental radiation. However, it is important to understand whether harmonic lasing provides significant benefits over the conventional method.

Studies that investigate harmonic lasing's potential to produce shorter wavelength radiation have used simulation results to justify the use of harmonic lasing in facilities such as the LCLS and the European XFEL and proposed various methods to suppress the growth of the fundamental harmonic to help the 3rd harmonic radiation grow [5] [7]. The fundamental harmonic must be suppressed because at saturation, the fundamental has harmful effects on the generation of the 3rd harmonic (this is due to the large energy spread of the electrons when they are bunched). This can be achieved by using phase shifters [5] [7]. One approach is to use phase shifters such that the 3rd harmonic gets shifted by a full 2π phase, while the fundamental is shifted by $2\pi/3$ [5]. This should shift the electrons off of resonance with respect to the fundamental and therefore disrupt the growth of the fundamental. The newly generated out of phase radiation should cancel with the old radiation and therefore suppress the fundamental radiation. Another method suggests using $2\pi/3$ and $4\pi/3$ phase shifts, one after the other [7]. Further study is needed to conclusively determine which way is best to suppress the fundamental. To further complicate matters, the fundamental harmonic is also useful to the growth of the 3rd due to the electron microbunching that takes place when the electrons interact with the radiation of the fundamental harmonic. Additionally, most studies conclude that the saturation power of the 3rd harmonic is about 1% of the fundamental, which is pretty low. Although studies have shown the possibility of using harmonic lasing to generate higher energy photons, further investigation is required before harmonic lasing can be used viably.

This study attempted to characterize the effect of suppressing the fundamental radiation on the growth of the third harmonic in the context of 3-D simulation code using the methods proposed by earlier studies [5] [7]. This study also

compared the generation of high energy radiation through optimizing for the third harmonic and through lasing at the fundamental. The simulation results, generated using Genesis 3-D simulation code, suggested that the phase shifters failed to completely suppress the fundamental radiation [6]. Furthermore, the most third harmonic power was generated when the fundamental was free to grow. Lastly, this optimal case was compared to generating the radiation as fundamental radiation. This comparison was done for several beam emittance and energy spread values.

2 Theoretical Considerations

The theoretical framework for the generation of higher harmonics and the effect that the phase shifters should have on the fundamental radiation and the third harmonic was analyzed. First, note that the undulator magnetic field is given by

$$B(z) = B_0 \cos(k_u z),$$

where B_0 is the field amplitude and k_u determines the undulator wavelength through $\lambda_u = 2\pi/k_u$. Consider the Free Electron Laser Resonance Condition given by

$$\lambda_r = \lambda_u \frac{1 + \frac{K^2}{2}}{2\gamma^2}, \tag{1}$$

where λ_r is the resonant radiation wavelength, λ_u is the undulator field period, γ is the electron beam energy normalized to mc^2, and K is the undulator parameter

$$K = \frac{eB_0}{mck_u} = 0.934\lambda_u[cm]B_0[T].$$

The electrons in the electron beam oscillate due to the undulator magnetic field. This causes the electrons to radiate. The radiation electric field then interacts with the electrons in a power exchange given by

$$\frac{d\gamma}{dt}mc^2 = -e\mathbf{v} \cdot \mathbf{E}, \tag{2}$$

where $\mathbf{v}$ is the velocity of the electron and $\mathbf{E}$ is the electric field of the radiation. Thus, when the electrons oscillate in the same direction as the electric field, they lose energy to the radiation. The goal is to make sure this power exchange continues. However, although the electrons are moving near c, the speed of light, the radiation travels at exactly c. Therefore, the best case scenario is that the electrons slip relative to the radiation by λ_r as they wiggle once, meaning that they traveled λ_u. Then, as the electrons wiggle once, they arrive in phase with the radiation, ready to continue the power exchange that drives the growth of radiation. This is exactly the requirement that the resonance condition satisfies. However, the electrons can also slip by $2\lambda_r$, $3\lambda_r$, etc. to arrive in phase with the

radiation. This is what gives rise to higher harmonics of radiation. The higher harmonics have wavelengths of

$$\lambda_n = \frac{\lambda_r}{n} \tag{3}$$

$$\text{for } n = 2,\ 3,\ 4, \text{ etc.}$$

These higher harmonics are significantly weaker than the fundamental. For example, the power from the even harmonics is nearly zero due to geometric considerations where there is an equal amount of power gain and loss and therefore no net power exchange. However the odd harmonics, the third in particular, can acquire a significant amount of power. The power is still much weaker than the power of the fundamental radiation. To maximize the amount of third harmonic power generated we must consider how the fundamental affects the third harmonic.

The fundamental radiation drives the bunching of the electrons. By interacting with the radiation, some electrons gain energy while others lose it. From classical mechanics, we know that particles with different energies travel different paths. In this case, the higher energy particles travel longer paths while those with less energy travel shorter paths. This leads to a longitudinal compression, also known as bunching. The electrons form these bunches at the fundamental radiation wavelength, and this drives exponential growth of both the fundamental and the third harmonic radiation. However, this bunching also leads to significant energy spread, since high and low energy electrons are now bunched together. Harmonic radiation is much more sensitive to the effects of a high energy spread, and thus, the fundamental also has a negative impact on the third harmonic [5] [7].

The resonance condition shows how the electrons can remain in phase with the radiation in order to maximize the power exchange between them. One way to disrupt the growth of the fundamental is simply to move the electrons off resonance with respect to the fundamental radiation. This is where phase shifters come in. Phase shifters are in drift sections between undulator sections. A chicane bending magnet is used so that the electron beam takes a longer path than the radiation which simply travels through. By the time the electrons arrive on the longitudinal axis once more, their phase with respect to the radiation has been changed. For example, a $2\pi/3$ phase shifter shifts the electrons a distance corresponding to $\lambda_r/3$, or one-third of the radiation period. However, since the third harmonic has a wavelength corresponding to $\lambda_r/3$, the electrons should arrive on resonance with respect to the third harmonic. This is the method that earlier studies have suggested to disrupt the growth of the fundamental while allowing the third harmonic to continue growing [5]. Theoretically, using phase shifters in this way should lead to complete cancellation of the fundamental after every three undulator sections.

Both energy spread and beam emittance lead to longitudinal dispersion. This is why higher harmonics are more sensitive to these effects - the requirement for resonance between the electron and the harmonic radiation is more

sensitive to variations in the longitudinal direction due to the high frequency and low wavelength of the harmonic. Therefore, higher harmonics put a more stringent requirement on σ_z, or the standard deviation of the longitudinal distribution of the electrons. We can relate the longitudinal dispersion to the energy spread that caused it with

$$\sigma_z = R_{56}\sigma_\gamma = 2\lambda_r N_u \sigma_\gamma, \tag{4}$$

where R_{56} is a transformation matrix coefficient relating an energy change to a change in z, σ_γ is the energy spread in terms of γ, the electron beam energy normalized to mc^2, λ_r is the fundamental radiation wavelength, and N_u is the number of undulator periods in an undulator section. Equation (4) only applies to the longitudinal dispersion generated over one undulator section. Now, to consider the longitudinal dispersion caused by the transverse beam emittance, we start by considering a small angular deviation x' where

$$x' = \frac{dx}{dz} = \frac{dx/dt}{dz/dt} = \frac{v_x}{v_z} = \tan\theta \approx \theta, \tag{5}$$

where the z-coordinate is the undulator axis, x is a transverse coordinate and θ is the angle between the z-axis and the transverse motion in the x-dimension. The small-angle approximation has been used. Notice that x' is indeed equivalent to a small angular deviation. Now consider one electron that travels straight along the undulator axis with speed v, and another that travels with an angular deviation of x' with speed v. The particle with the straight path travels a distance vt in the z-direction in time t while the particle with the angular trajectory travels a distance $vt\cos(x')$ in the z-direction. Thus the path difference can be written as

$$\Delta z = vt - vt\cos(x') = vt(1-\cos(x')) \approx vt(1-(1-\frac{(x')^2}{2})) = vt\frac{(x')^2}{2}, \tag{6}$$

where we have expanded the cosine into its second-order approximation. Now we rewrite our path difference as a longitudinal dispersion σ_z, our small angular deviation x' as σ_x' and our distance vt becomes $\lambda_u N_u$ as we are considering the dispersion over one undulator section. Equation (6) becomes

$$\sigma_z = \lambda_u N_u \frac{(\sigma_{x'})^2}{2}. \tag{7}$$

From basic beam optics, we can relate $\sigma_{x'}$ to the transverse beam emittance ϵ_x with

$$\sigma_{x'} = \sqrt{\frac{\epsilon_x}{\beta_x \gamma}}, \tag{8}$$

where β_x is the beta function. Solving equations (4), (7), and (8) as a system,

$$\epsilon_x = \frac{4\lambda_r \beta_x \gamma \sigma_\gamma}{\lambda_u}. \tag{9}$$

This result holds for ϵ_y as well, but β_y must be used in the calculation. Equation (9) is an important result. It allows us to treat energy spread as an equivalent emittance and vice-versa. This is important as we want to make sure that we see the natural dependence of harmonic power on the beam emittance and the energy spread. We do not want to work in regimes where one parameter dominates over the other. Thus, we used equation (9) to verify that our energy spread and emittance values were comparable. This was especially important when we were comparing generating 21 keV in the fundamental to generating 21 keV as the third harmonic of 7 keV.

3 Methods

In this study, the third harmonic was 21 keV radiation while the fundamental was 7 keV radiation. The electron beam used in simulations had an energy of 13.5 GeV. The transverse electron beam profile was Gaussian. When using the term emittance, the study refers to the transverse normalized beam emittance. The simulations are time-independent, unless otherwise noted. 1024 particles were simulated. The peak current was 4000 A for section 3.1, and 3000 A for the other sections (to simulate LCLS-II parameters more closely).

All results were obtained by using Genesis simulation code and Dr. Juhao Wu's Matlab scripts [6]. The scripts were used to optimize the taper and quadrupole profiles for a given set of input parameters. The code accomplishes the optimization by doing two iterations of a parameter by parameter scan. There are a total of 8 parameters that are scanned; 3 are for the taper profile and 5 are for the quadrupole focusing profile. The taper profile used in the code is of the form

$$K(z) = K_0 \left(1 - a \left[\frac{(z - z_0)}{(L_u - z_0)} \right]^b \right),$$

where K_0 is the initial undulator parameter, a is the taper ratio, b is the taper order, z_0 is the taper start point, and L_u is the total undulator length. The three parameters that the code scans over are a, b, and z_0. The code also scans over the quadrupole profile which is of the form

$$K_q(z) = \begin{cases} K_{q0}, & \text{for } 0 \leq z \leq z_1 \\ K_q(z_1)[1 - m_1(z - z_1)], & \text{for } z_1 \leq z \leq z_2 \\ K_q(z_2)[1 - m_2(z - z_2)], & \text{for } z_2 \leq z \leq L_u, \end{cases}$$

where K_{q0} is the initial quadrupole field strength, z_1 is the first quadrupole gradient variation starting point, m_1 is the middle quadrupole gradient, z_2 is the second quadrupole gradient variation starting point, m_2 is the final quadrupole gradient, and L_u is the total undulator length. The code scans over the five parameters K_{q0}, z_1, m_1, z_2, and m_2.

The code optimizes the taper and quadrupole profiles by scanning each parameter one by one. One scan consists of choosing several values in a specified range, using Genesis to run a simulation for each value, and choosing the best result. The best result was usually the one that resulted in the most third harmonic power; however, it was the value that resulted in the most fundamental power for the 21 keV as the fundamental case. The code does two such iterations of these parameter scans. Each run of the optimization code took about 5 hours, and every Genesis run varied from 30 seconds to 3 minutes, depending on parameter settings.

Genesis was also used to analyze the transverse radiation profile. Genesis produced one radiation profile after every undulator section. Thus, it was possible to monitor the evolution of the radiation profile in order to witness 3D effects that were taking place.

4 Results and Discussion

4.1 Implementing the phase shifters

In order to investigate the effect of using phase shifters, the fundamental power and third harmonic power were each plotted as a function of undulator distance for a variety of phase shifter schemes (see Figures 1 and 2). In each case, an optimization was conducted to tune the undulator parameters for optimal third harmonic power generation. Although the phase shifters resulted in frequent dips (as seen in Figure 1), the fundamental radiation continued to grow. Furthermore, each of the phase shifter schemes reduced the third harmonic power generated (see Figure 2). The alternating phase shifter scheme suggested by the authors of [7] seems to have produced the best results. Although it did not completely suppress the fundamental, the alternating scheme resulted in significant dips in the fundamental power. Indeed, the fundamental power is an order of magnitude lower than in the $2\pi/3$ case. Interestingly, the third harmonic of the alternating case is the same order of magnitude as the other phase shifter cases despite the lower fundamental power.

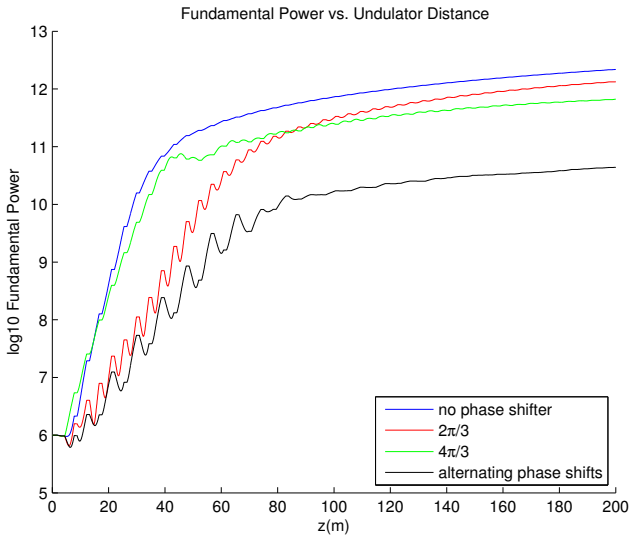


Figure 1: Fundamental power plotted as a function of undulator distance for no phase shift, $2\pi/3$, $4\pi/3$, and alternating phase shifts, for a normalized beam emittance value of $\epsilon = 0.3$ μm-rad, an energy spread $\sigma_\gamma = 2.54$, an undulator period of $\lambda_u = 0.032$ m, an undulator section length of $N_u = 106$ (in terms of the undulator period), and a total undulator length $L_u = 200$ m. Phase shifters were placed after every undulator section. The phase shifters did not successfully suppress the fundamental.

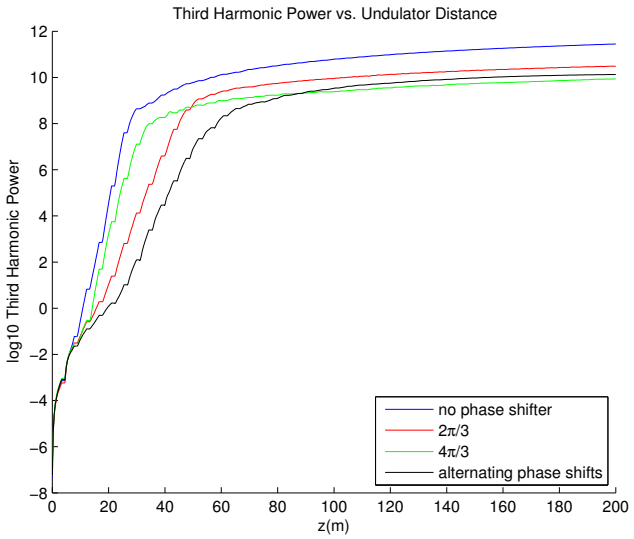


Figure 2: Third harmonic power plotted as a function of undulator distance for no phase shift, $2\pi/3$, $4\pi/3$, and alternating phase shifts, for a normalized beam emittance value of $\epsilon = 0.3$ μm-rad, an energy spread $\sigma_\gamma = 2.54$, an undulator period of $\lambda_u = 0.032$ m, an undulator section length of $N_u = 106$ (in terms of the undulator period), and a total undulator length $L_u = 200$ m. Phase shifters were placed after every undulator section. The phase shifter cases fared much worse than the no phase shifter case.

The fundamental power and the third harmonic power were also plotted as a function of beam emittance (see Figures 3 and 4). Once again, the taper and quadrupole profile was optimized for each case. The no phase shifter case did the best in terms of generating the most third harmonic power. This suggests that letting the fundamental grow unfettered is beneficial towards the third harmonic. The theory of harmonic generation corroborates these results as the fundamental does indeed drive bunching that is beneficial for the growth of the third harmonic. However, the phase shifters did not manage to suppress the fundamental; the power is still quite high.

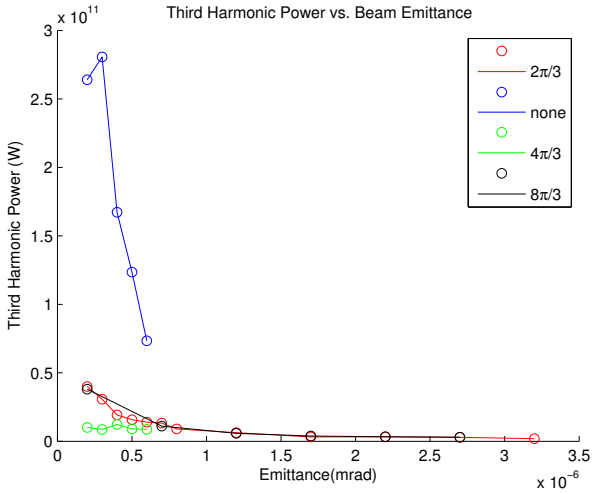


Figure 3: Third harmonic power plotted as a function of beam emittance for no phase shift, $2\pi/3$,$4\pi/3$, and $8\pi/3$. The phase shifter cases fared much worse than the no phase shifter case.

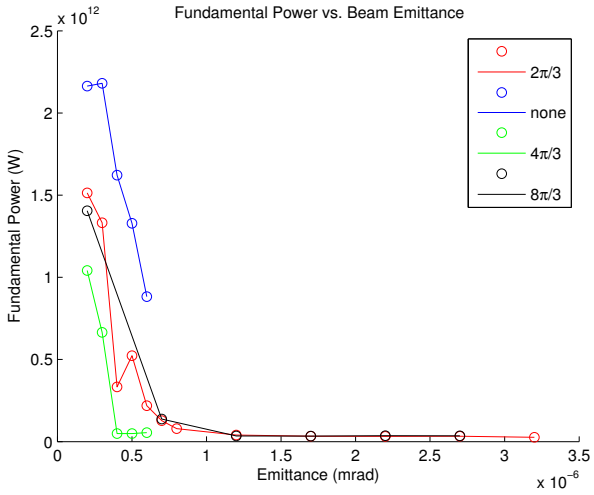


Figure 4: Fundamental power plotted as a function of emittance. Although the fundamental power appears to be quite low for some cases, this is due to high emittance values and the scaling of the plot. When cross-referenced with gain curves, it is evident that the fundamental is clearly not suppressed.

To verify this conclusion, the gain curves of the fundamental power were also plotted for some specific emittance values (see Figures 5 and 6). Although the $4\pi/3$ case appears to have suppressed the fundamental, a closer look at the gain curve reveals that proper suppression is not happening (see Figure 5). Consequently, most of the low power values in Figure 4 are due to a high emittance value (see Figure 6).

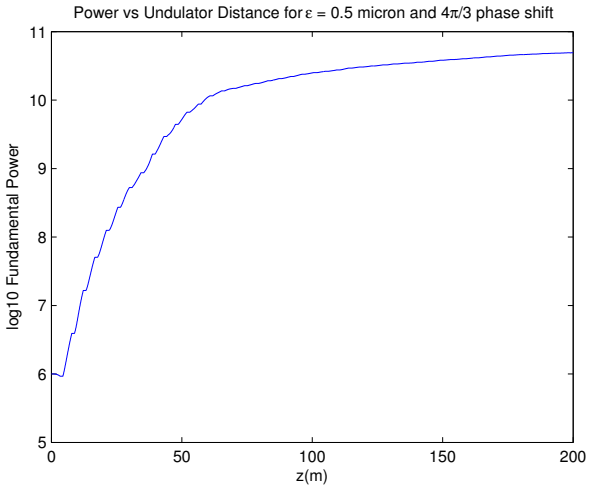


Figure 5: Fundamental power plotted as a function of undulator distance for an emittance of 0.5 microns and $4\pi/3$ phase shifters. The gain curve shows that the fundamental has not been suppressed.

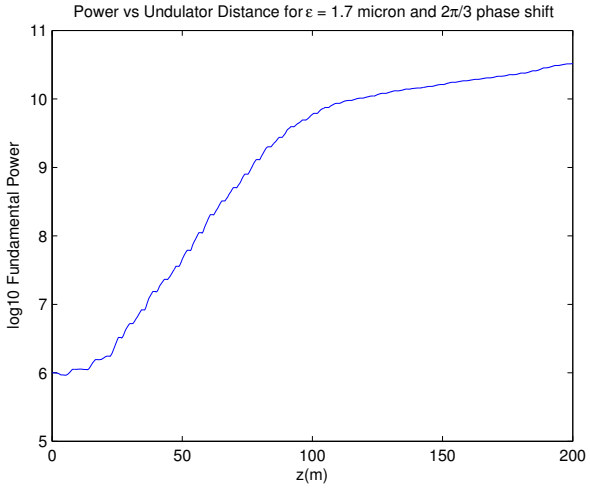


Figure 6: Fundamental power plotted as a function of undulator distance for an emittance of 1.7 microns and $2\pi/3$ phase shifters. The gain curve shows that the fundamental has not been suppressed. Rather, it is the high emittance value that is responsible for the low power at the end of the undulator.

In order to investigate whether other phase shifter values might be able to properly suppress the fundamental, simulations were conducted to generate plots of fundamental power and third harmonic power versus the phase shift angle (see Figures 7 and 8). Together, these plots support previous results, as the third harmonic power is highest when the fundamental power is also highest. None of the phase shifts managed to suppress the fundamental by a significant amount (see Figure 7). These results confirmed the hypothesis that there were more complicated reasons for why the fundamental was not being suppressed. One possible reason was the presence of 3-D effects such as diffraction. It was therefore necessary to explore the transverse radiation profile.

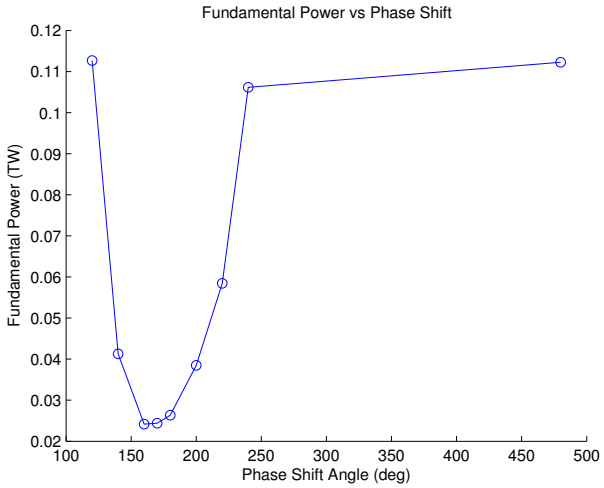


Figure 7: Fundamental power plotted as a function of phase shift. To isolate the phase shifter effect, the same taper profile and quadrupole profile was applied for each case (the optimal profile of the no phase shifter case).None of the phase shifts manage to significantly suppress the fundamental power.

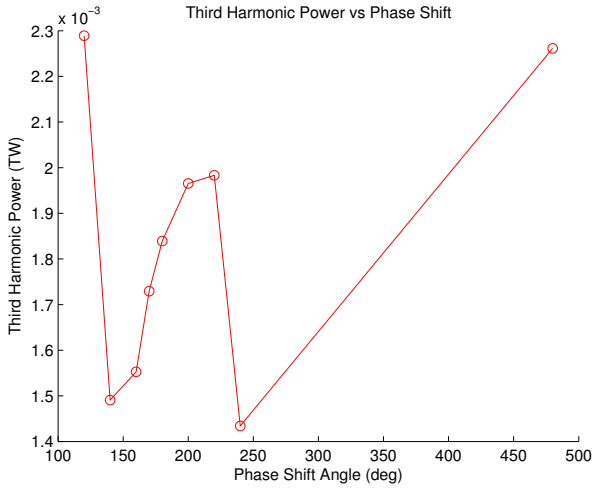


Figure 8: Third harmonic power plotted as a function of phase shift. Although there are some exceptions, it seems as though the largest third harmonic power values correspond to the largest fundamental powers.

4.2 Investigating the radiation profile

Since simulations confirmed that the phase shifters did not produce the expected cancellation of the fundamental radiation, we investigated the transverse radiation characteristics. To test whether there was a significant 3D diffraction effect that inhibited the interference that is critical to suppress the fundamental, we first implemented two phase shifts of π, one at 25 meters and one at 50 meters. In theory, the π phase shifts would cause significant cancellation and suppression. This would be due to the destructive interference of previously generated radiation with the newly generated π shifted radiation. However, if there was a significant diffraction effect taking place, there would be some cancellation near the origin of the xy plane but no effect for sufficiently large x and y values. If the previously generated radiation diffracted, the newly generated phase shifted radiation would only be able to interact and interfere with the previous radiation near the undulator axis, due to the radial expansion of the previous radiation.

To isolate the phase shifters' effect, the difference in radiation intensities between the no phase shifter case and the π phase shifter case was plotted at two locations - before the first phase shifter and after it (see Figure 9). Similar plots were generated for the difference in radiation phases (see Figure 10). There was no difference in radiation intensity before the first phase shifter, but after it, there were significant radiation intensity differences (see Figure 9). The colormap shows that the highest difference was near the origin, or the undulator axis, while there was little to no difference further away. The difference in intensity near the center of the radiation was about 10^6 times larger than the difference in intensity further away. This shows that far away from the center, the color map simply depicted noise. Consequently, the phase shifters affected the radiation that is in the center significantly, while the radiation that was far enough from the undulator axis was relatively unaffected.

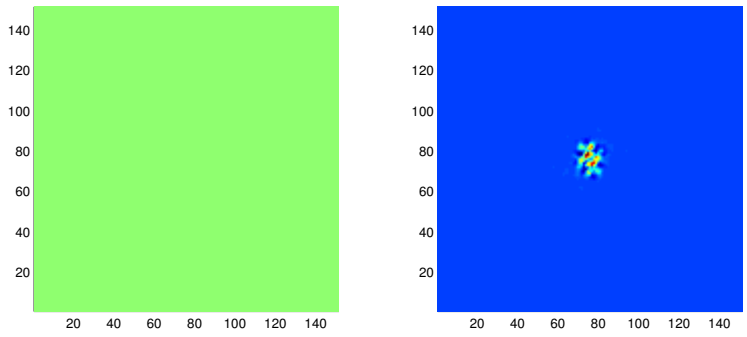


Figure 9: A color map showing the difference in radiation intensity between the no phase shifter case and the π phase shifter case as a function of the transverse xy plane. The left frame is before the first phase shift and the right frame is after it. Note that the same taper and quadrupole profiles were used in each case.

Similarly, there was no difference in phase before the first phase shifter. But after the π phase shifter, there was a constant phase difference in the center, and a few phase differences outside of the ring in the center. Although the phase shifters should shift the location of the electrons relative to the radiation, the radiation phase should be impacted when the new radiation interferes with the old radiation, as seen in the center of Figure 10. However, outside the center, there weren't significant phase differences. This shows that the phase shifters do not impact the outer regions of radiation very strongly. In fact, these results were sufficient to conclude that the phase shifters fail to provide the desired suppression due to the 3-D diffraction effects.

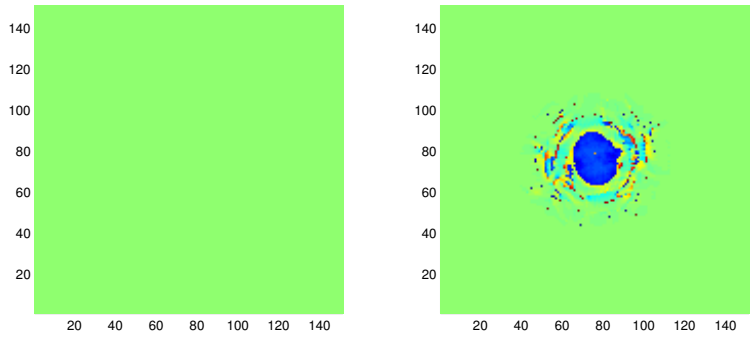


Figure 10: A color map showing the difference in radiation phase between the no phase shifter case and the π phase shifter case as a function of the transverse xy plane. The left frame is before the first phase shift and the right frame is after it. Note that the same taper and quadrupole profiles were used in each case.

The next logical step was to minimize diffraction effects in order to actually witness some suppression. Diffraction effects are negligible in the 1-D limit, and earlier studies that worked solely in the 1-D limit witnessed suppression [5]. Thus, it was prudent to try to reproduce those results by minimizing 3D effects.

4.3 Approaching the 1-D limit

The 1-D limit occurs when there is a transverse beam distribution that is like a pancake - the distribution in the transverse variables x and y has to be sufficiently larger than a gain length (the length it takes for the radiation power to grow by a factor of e). More formally,

$$Z_R \gg L_G, \qquad \text{(1-D Limit)}$$

where Z_R is the Rayleigh length and L_G is the gain length. From basic beam optics, the Gaussian beam has

$$Z_R = \frac{\pi \omega_0^2}{\lambda_r},$$

where ω_0 is the beam waist and λ_r is the radiation wavelength. The beam waist can be approximated as

$$\omega_0 \approx 2\sigma_x = 2\sqrt{\frac{\epsilon_{n,x}\beta}{\gamma}},$$

where σ_x is the standard deviation of the transverse distribution in x, ϵ_n, x is the normalized emittance in x, β is the beta function, and γ is the beam energy

normalized to mc^2. The x dimension was chosen without loss of generality; this analysis holds just as well for the y dimension. Substituting the approximation for the beam waist into the definition of the Rayleigh length,

$$\frac{4\pi\epsilon_{n,x}\beta}{\gamma\lambda_r} \gg L_G.$$

A typical gain length is roughly 2.2 meters, so a β that leads to $Z_R = 100$ was used, since $100 \gg 2.2$. The chosen β was 110 m. To run a simulation that more closely resembled the parameter settings in earlier studies, the length of each undulator section was halved so that it was approximately equal to a gain length, and the undulator length was set to be 25 undulator sections [5]. There was no tapering or optimization of any sort.

To find out if the phase shifters successfully suppressed the growth of the fundamental power, the fundamental power and third harmonic power were plotted as functions of undulator length for the phase shift and no phase shift cases (see Figures 11 and 12). Approaching the 1-D limit clearly produced the desired results - the fundamental was suppressed significantly by using the phase shifters. Thus, the phase shifter scheme manages to suppress the fundamental if beam parameters are close enough to the 1-D limit. However, earlier studies concluded that using the phase shifters suppresses the fundamental and allows the third harmonic to continue to grow, while here, the no phase shifter case had the largest third harmonic power (see Figure 12) [5]. Although fundamental was successfully suppressed, the results suggest that it may not be particularly prudent to do so. The LCLS-II does not run very close to the 1-D limit and furthermore, it is not possible to phase shift after every gain length. The LCLS-II has undulator sections that are about 3.4 meters long while the gain length for LCLS-II parameters is about 2.2 meters and phase shifters can only be placed in the gap sections between undulator modules. Thus, it is not possible to phase shift as often as required to suppress the growth of the fundamental.

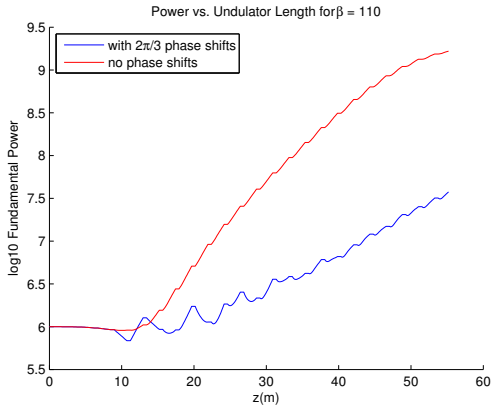


Figure 11: Fundamental power plotted as a function of undulator length for the phase shift and no phase shift cases with a beta function of 110 m. The phase shifters actually did manage to significantly suppress the fundamental, although not at the level predicted in [5].

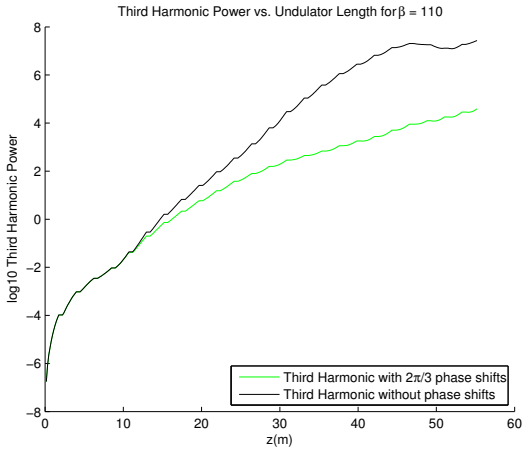


Figure 12: Third harmonic power plotted as a function of undulator length for the phase shift and no phase shift cases with a beta function of 110 m. Contrary to what was found in [5], the third harmonic power is still higher when there are no phase shifters.

Although it was not possible to further optimize the generation of the third harmonic by using phase shifters, it was still desirable to compare generating higher-energy radiation through the fundamental to generating the radiation

through harmonic lasing as the third harmonic.

4.4 Harmonic generation versus the fundamental

In order to further investigate the feasibility of harmonic lasing, we compared generating 21 keV radiation as the third harmonic of 7 keV radiation to generating 21 keV radiation in the fundamental. To do so, the radiation power in each case was plotted as a function of beam emittance in order to see if beam emittance factored in to preferring one method of generation over another (see Figures 13 and 14). The energy spread value used was $\gamma\sigma_\gamma = 1$, where γ and σ_γ are consistent with equation (9), in order to ensure a fair comparison between the two cases. Note that the simulations in Figure 13 were run with no tapering of the magnetic lattice while every simulation run used to produce Figure 14 had an optimized tapering profile. Figure 17 demonstrates that generating the 21 keV as the fundamental is preferred to generating it as the third harmonic in terms of maximum power, when there is no tapering of the magnetic lattice. However, the power in both cases become nearly identical in a higher emittance regime. Figure 18 demonstrates that the powers for each case become much more comparable once the taper optimization is carried out, especially for beam emittance values greater than 0.2 microns. Furthermore, there is a correlation here between the 21 keV third harmonic power and the 7 keV fundamental. As seen earlier, higher third harmonic power is a result of higher fundamental power.

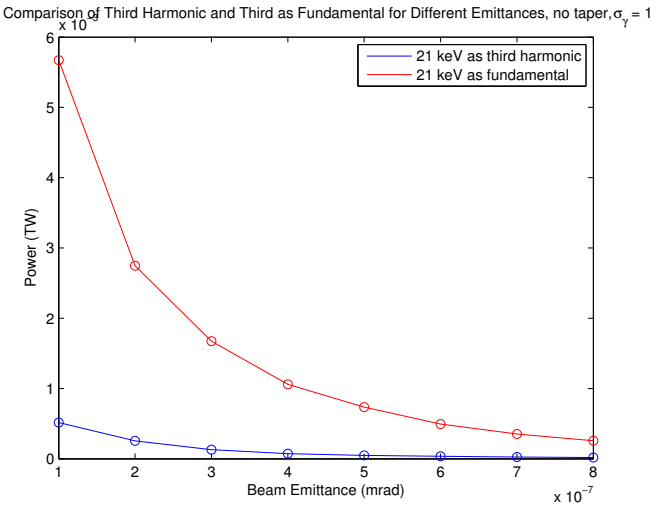


Figure 13: Comparison of generating 21 keV as the third harmonic to 21 keV as the fundamental as a function of beam emittance. This is the no taper case. The exponent on the y-axis is 10^-3.

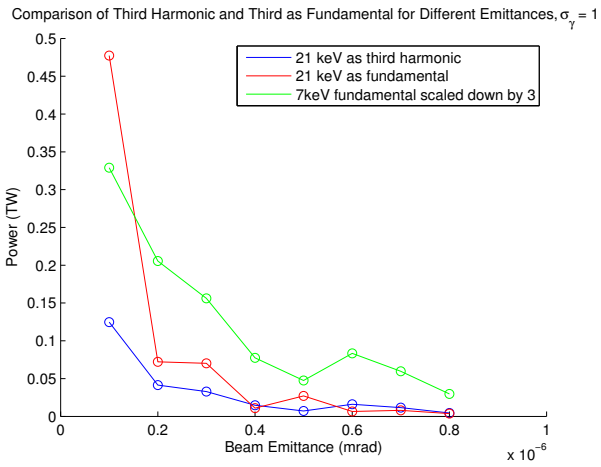


Figure 14: Comparison of generating 21 keV as the third harmonic to 21 keV as the fundamental as a function of beam emittance. The taper was optimized for every case. The 7 keV fundamental was also included, although it is scaled down by a factor of 3.

5 Conclusion

Harmonic lasing has the potential to be a very effective tool when trying to generate short wavelength, high energy radiation; however, much more investigation is needed. Future work would focus on developing a more complete theory on the optimization of the third harmonic. This includes understanding how to suppress the fundamental properly and whether it is actually beneficial to do so. Even though this study identified a few problems with suppression such as diffraction, further study into these problems and potential solutions could yield valuable results. Based on the results of this study, a future study could try increasing the number of phase shifters along the entire undulator, or find a more promising alternating phase shifter scheme. It is equally important to explore the merit of suppressing the fundamental. Even when the fundamental was suppressed in the 1-D limit, the third harmonic power was lower than that of the no phase shifter case. Further investigation into the physics of harmonic generation would be required to explain this in a more precise manner. There might be a regime in which the suppression of the fundamental allows for proper optimization of the third harmonic, as the authors of [5] suggest. The results of this study imply that to generate the most third harmonic power, the fundamental needs to grow unfettered. This study also showed that harmonic lasing is comparable to generating higher energy radiation in the fundamental. Future work could also focus on a different optimization scheme for third harmonic power (like the phase shifter scheme, but something that works in the

3-D limit).

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7 Acknowledgements

I would like to acknowledge both Professor Tor Raubenheimer and Dr. Juhao Wu for their outstanding support and valuable insight. Discussions with Tor and Juhao were critical to the completion of this work, and extremely beneficial to me as I worked towards developing a thorough understanding of the underlying physics. I would also like to thank Dr. Xiaobiao Huang and Dr. Lanfa Wang for their help and guidance. I would also like to thank Claudio Emma for his clear and concise explanations of basic FEL physics as well as my cubemate Ryan Roussel for always being ready to lend a helping hand. Finally I would like to thank my reviewers - Professor Adam Wierman, Arun Chandra, John Naruk, Ritvik Mishra, and Gillian Pierce for helping me refine the writing in this paper.